EEG-Based Machine Learning Analysis for ssVEP Task Classification and Neural Feature Prediction

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Basic Information

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Abstract

This project presents a machine learning pipeline for analyzing electroencephalography (EEG) recordings from a steady-state visually evoked potential (ssVEP)-style brain-computer interface dataset. The objective was to evaluate whether EEG frequency-band features could classify experimental task intent and predict neurophysiological response measures such as steady-state time and an occipital P100 latency proxy. Clinical metadata from an Excel sheet was fused with raw EEG CSV files, producing a master dataset of 1,236,488 samples from 138 sessions and 29 subjects. The EEG signals were segmented into 1-second epochs at 128 Hz, and Alpha, Beta, and Gamma band-power features were extracted from 14 electrodes using Welch's power spectral density method. Machine learning models were evaluated using subject-grouped cross-validation to prevent data leakage between training and testing sets.

The best five-class task classifier was a Random Forest model, achieving 27.86% accuracy and 0.2609 macro F1-score compared with a 22.64% majority-class baseline. Regression analysis initially suggested strong prediction of steady-state time, but target auditing showed that this variable was mostly fixed by task category, making it unsuitable as a continuous cognitive-agility target. P100 proxy latency regression did not outperform baseline models. Interpretability analysis showed distributed Beta and Gamma band contributions rather than a clean occipital visual-cortex signature. Overall, the project demonstrates a complete EEG machine learning workflow while highlighting the importance of leakage-safe validation, target auditing, and biologically specific ssVEP feature extraction.

Keywords: EEG, ssVEP, Machine Learning, Brain-Computer Interface, Signal Processing

1 Introduction

Brain-computer interfaces (BCIs) are systems that attempt to decode brain activity and convert it into useful commands or measurements. EEG is commonly used in BCI research because it is non-invasive, relatively low-cost, and capable of recording brain electrical activity with high temporal resolution [1, 3].

Steady-state visually evoked potentials (ssVEPs) are EEG responses generated when a person observes a visual stimulus flickering at a constant frequency. In a well-designed ssVEP system, the visual cortex produces activity at the stimulus frequency and its harmonics. These responses are often strongest over occipital electrodes such as O1 and O2, which are located near visual-processing regions [2].

The P100 waveform is another visually relevant EEG component. It is a positive deflection that typically appears approximately 100 ms after a visual stimulus. Clinically, P100 latency can be useful for studying visual pathway function [4].However, true P100 analysis requires epochs that are time-locked to visual stimulus onset. In this project, because event markers were not used for stimulus-locked averaging, the extracted P100-related feature is treated as a P100 latency proxy rather than a clinically definitive P100 measurement.

The original aim of this project was to test three machine learning tasks:

Task 1: Intent classification by predicting the experimental task name from EEG features.

Task 2: Cognitive agility regression by predicting steady-state time.

Task 3: Neural health regression by predicting an occipital P100 latency proxy.

The project also investigated whether demographic variables such as age and gender, and visual electrodes such as O1 and O2, contributed meaningfully to model predictions.

2 Proposed Methodology

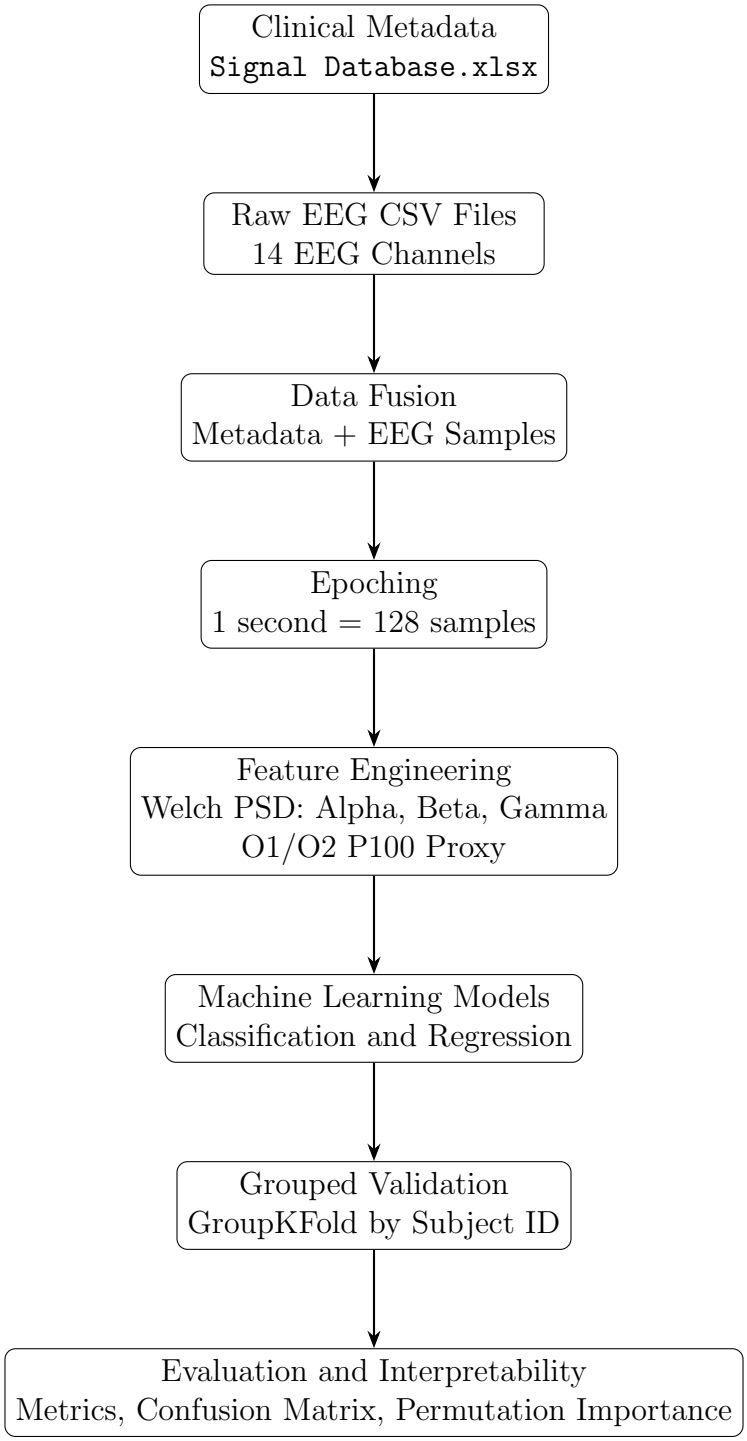

Figure 1: Proposed EEG machine learning pipeline from metadata ingestion to model evaluation and interpretability.

2.1 Tools and Libraries Used

The implementation was performed in Python using pandas and NumPy for data handling, SciPy for signal processing, scikit-learn for machine learning and validation, XGBoost for gradient boosting, and Matplotlib/Seaborn for visualization.

2.2 Dataset

The data set consisted of clinical metadata stored in `Signal Database.xlsx` and raw EEG recordings stored as CSV files. Each EEG file contained 16 columns, including a counter column, an interpolated column, and 14 EEG electrode channels.

The 14 EEG channels used were:

`AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4`

During data ingestion, metadata such as age, gender, test name, test type, and steady-state time was attached to every EEG sample in the corresponding recording.

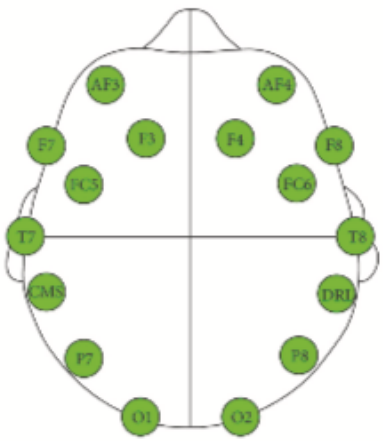

FIGURE 4: International 10-20 system for Emotiv Epoc+.

Figure 2: Layout for an Emotiv EPOC+ headset. Source: dataset documentation.

Table 1: Data ingestion summary

Item	Value
Metadata-listed sessions	141
Successfully processed sessions	138
Unique subjects	29
Recovered filename mismatches	21
Truly missing sessions	3
Final master dataset rows	1,236,488
Final master dataset columns	25

Three files were unavailable and excluded: `A006SV1_1.csv`, `A012SV1_2.csv`, and `A020SV1_2.csv`. To preserve auditability, the final dataset stored both the expected file name and the actual file name used when filename mismatches were resolved.

Description of Experimental Tasks

The dataset contains multiple experimental task labels under `Test_Name`. These task labels represent the condition being performed during each EEG recording.

- **FIVE BOX 1, FIVE BOX 2, FIVE BOX 3:** These tasks appear to be visually similar five-box stimulus conditions. They likely involve attention to visual targets arranged in box-like regions. Because the three tasks are visually similar, their EEG patterns may overlap, which explains why the classifier often confused them.
- **IMAGE SEARCH:** This task likely involves visually searching for or identifying an image-based target. In the classification results, this was the most recognizable class, suggesting it may produce a more distinct visual-attention pattern than the other tasks.
- **HAND SHAKE:** This task appears to involve a motor-imagery or hand-movement-related condition. It was the weakest class in the classifier, possibly because broad Alpha/Beta/Gamma power features did not capture its specific neural pattern clearly.

Because detailed task protocol documentation was limited, these descriptions are based on the task names and observed model behavior. Therefore, they should be treated as practical interpretations rather than confirmed experimental definitions.

2.3 Pre-processing

Each raw CSV file was loaded by skipping the first row and manually assigning the EEG column names. The `Interpolated` column was removed because it contained no useful signal information. The data was grouped by session so that each recording could be processed independently.

The sampling frequency of the EEG headset was 128 Hz. Therefore, one second of EEG data corresponded to 128 samples. The continuous EEG data was divided into 1-second epochs. This produced an epoch-level dataset suitable for machine learning.

2.4 Feature Engineering

Frequency-domain features were extracted using Welch's power spectral density (PSD) method. Welch's method estimates the distribution of signal power across frequencies and is commonly used in EEG analysis because it reduces noise compared with a raw Fourier transform [5].

For each 1-second epoch, average power was extracted from three standard EEG frequency bands:

Table 2: Frequency bands used for feature extraction

Band	**Frequency Range**
Alpha	8–12 Hz
Beta	12–30 Hz
Gamma	30–60 Hz

These band-power features were extracted for all 14 EEG electrodes, producing 42 main EEG frequency features per epoch.

For the P100 proxy feature, the EEG signal was filtered between 1 and 30 Hz, and the highest positive peak between 80 ms and 120 ms was extracted from the O1 and O2 electrodes. This was recorded as an occipital P100-style latency proxy. Since the data was not stimulus-locked, this feature was not interpreted as a clinical P100 measurement.

2.5 Machine Learning Models

The following models were evaluated for intent classification:

- Dummy majority-class baseline
- Logistic Regression
- K-Nearest Neighbors
- Support Vector Machine with RBF kernel
- Random Forest Classifier
- XGBoost Classifier

These models were selected because they represent common linear, distance-based, kernel-based, ensemble, and boosting approaches used in EEG and general machine learning classification [6, 7, 8, 9]. For regression tasks, the following models were evaluated:

- Dummy mean baseline
- Linear Regression
- Ridge Regression
- Random Forest Regressor

2.6 Evaluation Strategy

The dataset contained many 1-second epochs from the same subjects. A normal random train-test split would allow epochs from the same subject to appear in both training and testing sets, causing data leakage. To avoid this, all model evaluation used grouped validation with `Subject_ID` as the group variable.

This means that the models were tested on subjects not seen during training, making the evaluation more realistic and scientifically defensible.

2.7 Explanation of Key Methods

Welch's Power Spectral Density Method: Welch's method is a signal-processing technique used to estimate how much power a signal contains at different frequencies. In EEG analysis, this is useful because brain activity is often studied through frequency bands such as Alpha, Beta, and Gamma. Instead of using raw voltage values directly, Welch's method converts each EEG epoch into frequency-power features that are easier for machine learning models to use.

Welch's T-Test: Welch's t-test is a statistical test used to compare the means of two groups. It is useful when the two groups may have different variances or unequal sample sizes. In this project, it was used to compare male and female subject-level P100 proxy latency.

Mann-Whitney U Test: The Mann-Whitney U test is a non-parametric test used to compare two groups without assuming that the data follows a normal distribution. It was used as a robust check for the gender comparison because the sample size was small.

GroupKFold Cross-Validation: GroupKFold is a validation method that keeps all data from the same subject together in either the training set or the testing set. This is important for EEG datasets because each subject contributes many epochs. Without grouping, the model could memorize subject-specific patterns and produce overly optimistic results.

Macro F1-Score: Macro F1-score calculates the F1-score separately for each class and then averages them equally. This is useful when classes are imbalanced because it prevents the majority class from dominating the evaluation.

Permutation Importance: Permutation importance is an interpretability method that measures how much model performance decreases when one feature is randomly shuffled. If shuffling a feature greatly reduces performance, that feature is considered important for the model.

3 Result and Discussion

3.1 Exploratory Data Analysis

Exploratory analysis showed a possible upward trend between age and steady-state time. However, the confidence interval was wide, so this relationship was treated as exploratory rather than conclusive.

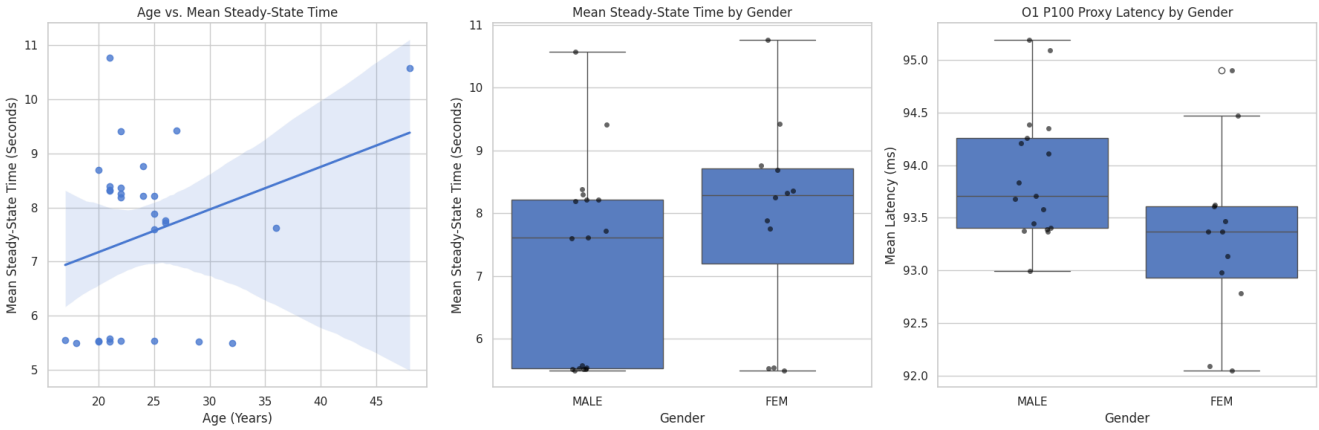

Figure 3: Demographic and target distribution analysis.

A subject-level gender analysis was performed for the O1 P100 proxy latency. After normalizing gender labels, there were 17 male subjects and 12 female subjects. Male subjects had a mean O1 P100 proxy latency of 93.91 ms, while female subjects had a mean latency of 93.32 ms.

Table 3: Gender comparison for O1 P100 proxy latency

Metric	**Value**
Male mean latency	93.91 ms
Female mean latency	93.32 ms
Welch's t-test p-value	0.0524
Mann-Whitney U p-value	0.0395

The Welch's t-test narrowly missed the conventional 0.05 significance threshold, while the Mann-Whitney U test suggested a significant distributional difference. Due to the small sample size and proxy nature of the P100 feature, this result was interpreted as suggestive rather than conclusive.

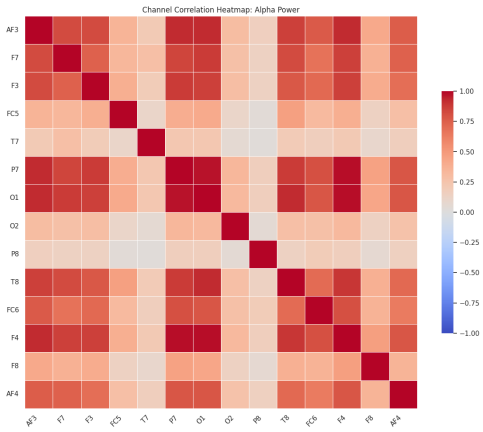

Figure 4: Channel Correlation Heatmap

The Alpha-band correlation heatmap showed broad positive correlations across EEG channels. This indicated multicollinearity among EEG predictors and justified testing regularized models such as Ridge Regression.

3.2 Task 1: Intent Classification

The first machine learning task was to predict `Test_Name` from one second of EEG frequency-band features.

Table 4: Intent classification model comparison

Model	**Accuracy**	**Macro F1**	**Weighted F1**
Random Forest	0.2786	0.2609	0.2680
XGBoost	0.2557	0.2448	0.2497
Logistic Regression	0.2562	0.2400	0.2432
KNN	0.2267	0.2221	0.2263
SVM RBF	0.2257	0.1664	0.1697
Dummy Majority Baseline	0.2264	0.0738	0.0836

The best model was the Random Forest classifier, with 27.86% accuracy and 0.2609 macro F1-score. The majority baseline achieved 22.64% accuracy but only 0.0738 macro F1-score. This shows that the classifier learned some task-related information, but the performance was still weak.

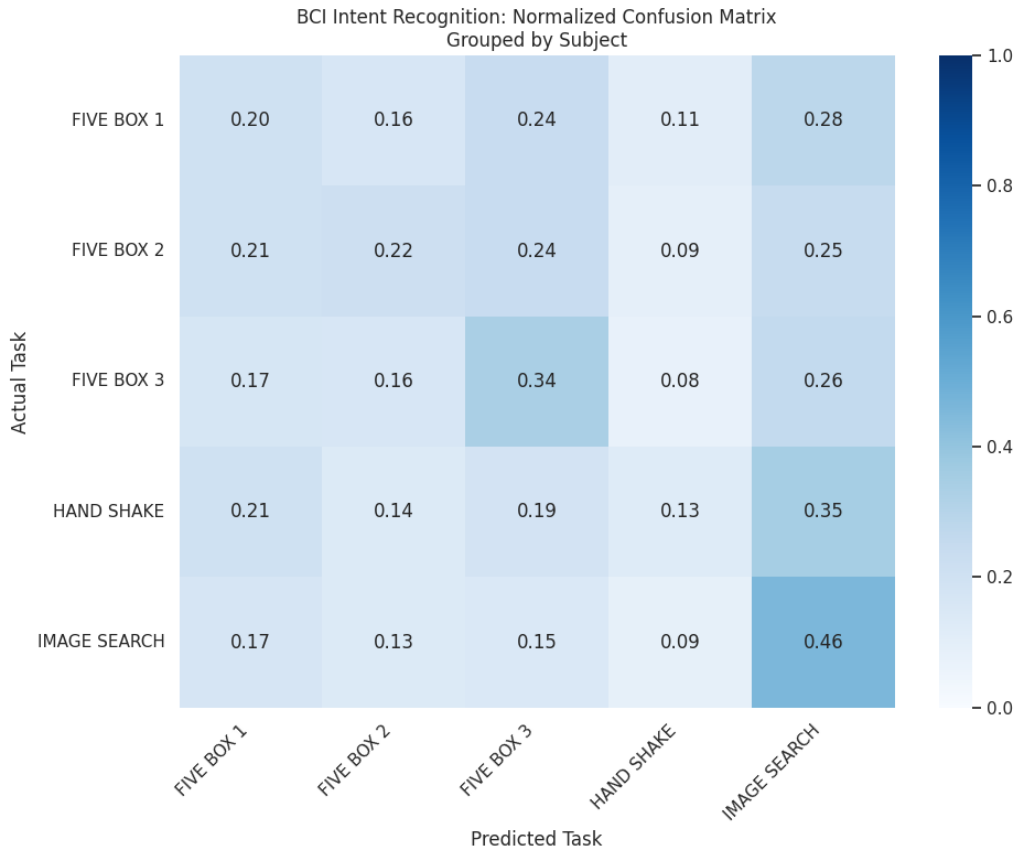

Figure 5: Normalized confusion matrix for five-class intent classification using subject-grouped validation.

The confusion matrix showed that `IMAGE SEARCH` was the most recognizable class, while `HAND SHAKE` was the weakest. The three `FIVE BOX` tasks were frequently confused with one another, which is reasonable because they were visually similar tasks.

3.3 Task 2: Cognitive Agility Regression

The second task was to predict `Steady_State_Time`. Initially, Random Forest regression appeared to perform well, achieving an R^2 value of 0.6931. However, a target audit showed that steady-state time was mostly fixed by task category.

Table 5: Steady-state time target audit

Task	Unique Values	Minimum	Maximum	Standard Deviation
FIVE BOX 1	1	5	5	0
FIVE BOX 2	1	5	5	0
FIVE BOX 3	1	5	5	0
IMAGE SEARCH	1	5	5	0
HAND SHAKE	2	9	25	7.3533

This revealed that `Steady_State_Time` was not a valid continuous cognitive-agility

target across all tasks. The model was mainly learning task structure rather than predicting a true subject-level cognitive response measure.

3.4 Task 3: Neural Health Regression

The third task was to predict `O1_P100_Proxy_Latency_ms`. The models did not outperform the dummy baseline.

Table 6: P100 proxy latency regression results

Model	**RMSE**	**MAE**	R^2
Dummy Mean	12.7090	11.3940	-0.0002
Random Forest Regressor	12.7905	11.5369	-0.0131
Ridge Regression	13.3168	11.5515	-0.0981
Linear Regression	16.2023	11.6505	-0.6256

This result indicates that the available broad-band EEG features did not reliably predict the P100 proxy latency. Additionally, because the P100 feature was not stimulus-locked, it should not be interpreted as a clinical neural-health marker.

3.5 Model Interpretability

Permutation importance was used to inspect model behavior. For steady-state time regression, `Test_Name` was by far the strongest predictor, confirming that the target was mainly task-defined. When `Test_Name` was removed, the model failed to generalize, with a negative R^2 value.

For the Random Forest intent classifier, the most important features were mainly Beta and Gamma band-power features. The strongest channel-level contributors were FC5, T8, T7, FC6, and P7. O1 contributed modestly, but O2 did not emerge as an important feature.

Table 7: Intent classifier frequency-band importance

Frequency Band	**Total Importance**
Beta	0.0453
Gamma	0.0446
Alpha	-0.0278

The interpretability results did not support the claim that the model isolated the visual cortex through O1 and O2. Instead, the model relied on distributed features across temporal, frontocentral, and parietal regions. This suggests that broad Alpha, Beta, and Gamma features were not specific enough for clean ssVEP decoding.

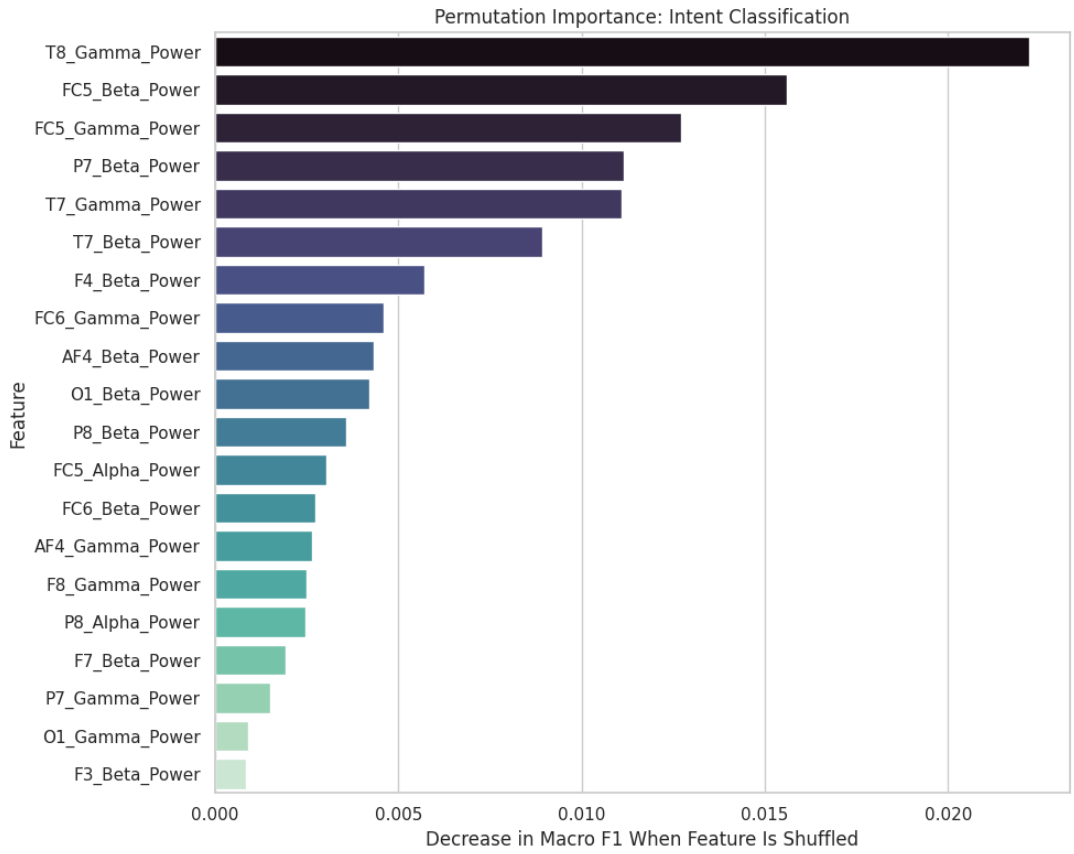

Figure 6: Permutation importance of EEG band-power features for the Random Forest intent classifier.

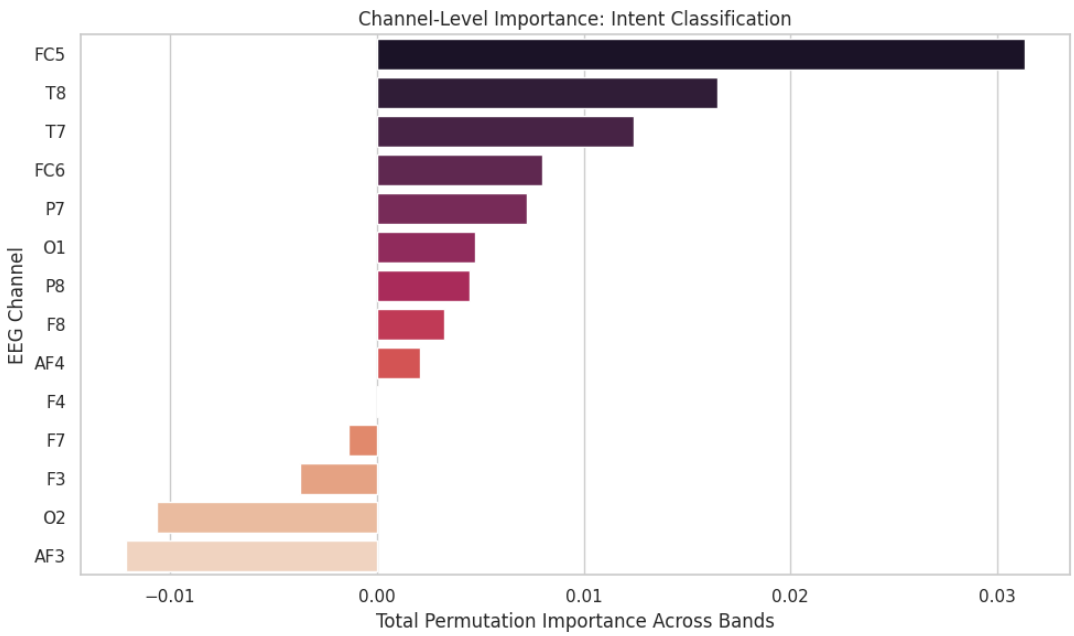

Figure 7: Channel-level importance for intent classification, summed across Alpha, Beta, and Gamma bands.

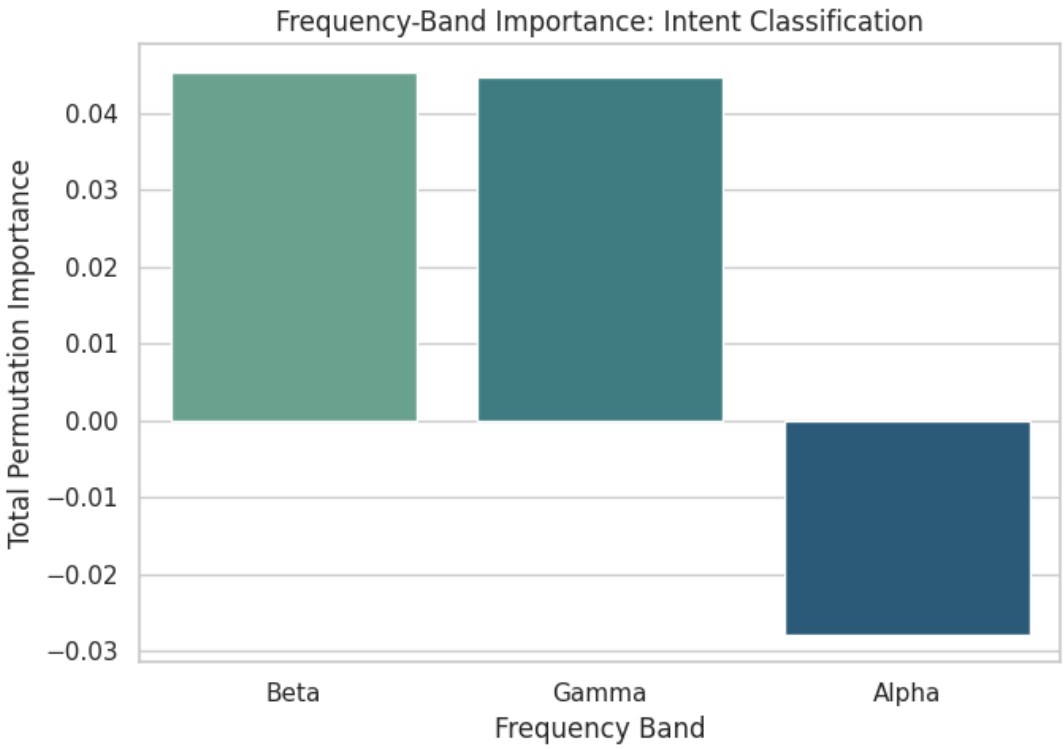

Figure 8: Frequency Band Importance of Alpha, Beta, and Gamma bands.

4 Conclusion and Future Work

This project successfully implemented a complete EEG machine learning pipeline, including metadata fusion, EEG preprocessing, epoching, frequency-domain feature extraction, grouped model validation, regression analysis, and interpretability.

The five-class intent classification task showed modest above-baseline performance. The best Random Forest model achieved 27.86% accuracy compared with a 22.64% majority-class baseline. However, this was not strong enough to claim a reliable real-world BCI decoder.

The steady-state time regression task was found to be unsuitable as a general cognitive-agility prediction problem because the target value was mostly fixed by task category. The P100 proxy regression task also did not produce meaningful predictive performance.

The main conclusion is that the machine learning pipeline was technically successful, but the dataset and broad-band feature set limited the strength of the biological conclusions. The results show the importance of checking target validity and preventing subject-level data leakage in EEG machine learning.

The main limitations of this study were the absence of stimulus event markers, the use of broad frequency bands instead of ssVEP stimulus-frequency features, the small number of subjects, and the limited validity of Steady-State Time as a continuous regression target.

Future work should include:

- Using stimulus event markers to create true stimulus-locked epochs.
- Extracting ssVEP-specific stimulus-frequency and harmonic power features.
- Focusing on occipital electrodes O1 and O2 for visual-response analysis.
- Collecting more subjects and more balanced task labels.
- Using clinically validated labels for neural-health prediction.
- Testing deep learning or time-series models only after improving feature quality.

Overall, this project should be understood as a rigorous EEG machine learning feasibility study. It demonstrates that broad-band EEG features alone are insufficient for strong subject-generalized BCI decoding on this dataset, and it identifies clear directions for improving future ssVEP-based models.

Explanation Notes

What did this project do? This project converted raw EEG recordings into machine learning features and tested whether those features could predict task intent, steady-state time, and a P100-style latency proxy.

Why was GroupKFold used? GroupKFold was used because there were many epochs from the same subject. Without grouping, the same subject could appear in both training and testing data, causing data leakage.

Did the classifier work? It worked weakly. Random Forest performed better than the majority baseline, but the performance was not high enough for a reliable BCI.

Why did steady-state regression fail as a scientific target? The target was mostly fixed by task. Four tasks always had a steady-state time of 5 seconds, so the model was learning task identity rather than cognitive speed.

Why is P100 called a proxy? True P100 requires EEG epochs aligned to visual stimulus onset. Since event markers were not used, the extracted 80–120 ms peak is only a proxy.

What is the final conclusion? The pipeline was successful, but the dataset and broad-band features were not sufficient for strong BCI or clinical claims.

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